Repair of complete denture

Complete dentures are predominantly fabricated from acrylic resin, which is relatively weak and prone to fatigue and impact fracture.

Causes of fracture:

1. Impact fracture usually results from dropping the denture either onto the floor or into a sink while cleaning the prosthesis.
2. fabrication errors include excessive thinning of the palate of the upper denture.
3. Excessively accentuated frenal notches that act as stress raising features, resulting in crack initiation and subsequent propagation.
4. Excessive loads typically arise when a complete denture is opposed by natural teeth.

Types of repair:

1-maxillary denture base fracture all of the parts are available

2-mandibular denture base fracture all of the parts are available

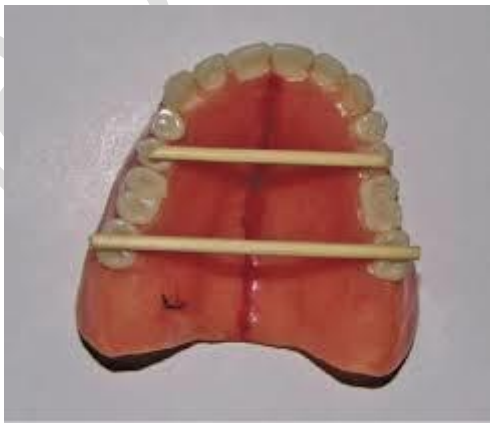
3-replacement of the broken or missing tooth or teeth

4-missing labial or lingual border

Dr. Wasmaa Sadik

1-maxillary denture base fracture all of the parts are available

1. Assemble the broken pieces of denture carefully and secure with sticky wax. It may be necessary to strengthen the joint by securing an old bur across the teeth with sticky wax. On the fitting surface, block out with wax any large undercuts that do not involve the fracture site to enable easy removal of the model later.
2. Cast a model to the fitting surface of the denture using plaster mixed after placing separating medium on the tissue side of the denture.



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3. When set, remove the denture from the new model, and then remove the wax from the denture.

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4. Reduce the broken ends slightly to expose clean acrylic resin, and create a definite gap between the two parts of the denture when assembled on the model. A finishing line for the repair resin to 'butt' up to can also be ground into the resin.

5. Paint the model with separating medium, position the denture on the model and secure with sticky wax.
6. Mix the acrylic and fill the gap, leaving a slight excess.

7. Place in a water pressure bath and cure for 15 minutes at 3 atm of pressure at 45°C.

8. After curing and removal of the denture from the model trim the cured material to leave the original contour of the denture.
9. Smooth with sandpaper and polish

Dr. Wismaa Sa

Replacement of the broken or missing tooth or teeth

Fracture of a single tooth may involve the entire tooth or just a part of it may fracture, leaving a fragment attached to the denture base. If the entire tooth has come off the denture, this is usually due to poor bonding with the acrylic resin (usually caused by separating medium left on the tooth during processing). If the tooth has fractured and a part of the tooth left bonded to the denture base, there may be a problem with the occlusion and this should be checked and corrected after placing the new tooth.

In the following example a lower incisor was damaged during the processing of a denture.

1. Using a tungsten bur, remove the tooth from the denture.
2. Enlarge the tooth socket to allow room for new acrylic.
3. Position the new tooth in place using wax.
4. Mix a small amount of plaster using anti-expansion solution and make a plaster key to fit around the tooth, wax work and adjacent teeth and allow to set. Alternatively, use silicone putty.
5. Boil off the wax from the denture and plaster key.
6. Assemble the parts and secure with sticky wax.
7. Mix a small amount of self-curing acrylic to a runny consistency and fill the void. Place in a water pressure bath and cure for 15 minutes at 3 atm of pressure at 45°C.
8. Remove the plaster key and trim the denture to its original contour.
9. Polish the repair

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Missing labial or lingual border:

– An impression is made with the denture placed in patient mouth.

– After pouring the cast; either self cure is applied to replace the missing part, or wax is added and carved to resemble the broken denture part followed by flasking, packing, curing, finishing and polishing.